

Question number	Answer	Additional guidance	Mark
1)	An answer that includes three of the following points: • toxicant absorbed into the blood (in the gut) (1) • crosses blood brain barrier (1) • {prevents production/less secretion} of dopamine (1) • inhibits uptake of dopamine by motor control centres of the brain (1)	accept killing dopamine producing cells	(3)

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2)	An answer that includes two of the following points: • use microarray (to identify affected gene(in sample) (1) • technique detail (1) • use of bioinformatics / algorithms (to analyse the data from the microarray (1)	E.g. cut using endonuclease enzymes / use of fluorescent dyes any reasonable technique PCR, amplification accept reference to use of computer database	(2)

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3)	An explanation that includes three of the following points: • process to {sequence genes/ DNA fingerprinting/DNA profiling} (1) • compare sequences to show similarities between (species/ humans and zebrafish) (1) • show product of the genes have (same/similar/related) functions (1)	accept reference to amino acid sequence of protein Accept use of PCR to make multiple copies comparing bands (only in context PCR)	(3)

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8(d)	An answer that includes the following points: {target/ bind to} transcription factor - they could change the shape of the transcription factor (1) - they could prevent binding of transcription factor to DNA (1) - RNA polymerase unable to bind (1)	accept damage to transcription factor accept prevent binding to receptors / promoters accept inactivates the transcription factor	Choose item. (3)

Question number	Answer	Additional guidance	Mark
8(e)	An answer that includes the following points: (breaks in the strands due to) phosphodiester bonds being broken (1) (strands are separated due to) hydrogen bonds being broken (1) - causes mutation (1)		Choose item. (3)

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(f)	A description that includes four of the following points: - use of {living / viable cells}(1) - reduced NAD produced during aerobic respiration(1) - use the same concentrations of drugs (1) - record how long it takes the solution to turn purple / accept reference to use of colorimeter(1) - the longer it takes to turn purple the more toxic the drug (1)	accept converse	(4)

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(g)	An answer that includes two of the following points: • individuals have different genetic background / make up (1) • because response to toxicity is individual(1) • because embryonic stem cells do not have to be used so no ethical issue (1)	assays takes into account individual epigenetic influences no destruction of embryo so no ethical issues	(2)